

A Shared Intelligence Network for Employment-Experience Fraud: Privacy-Preserving Signal Sharing and Incentive Design

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Abstract

Misrepresentation of professional experience—fabricated seniority, invented employment timelines, and false credentials—is one of the most common and least-policed forms of hiring fraud, and generative models that mass-produce convincing synthetic candidates are lowering its cost to near zero. Two mature bodies of work bear on the problem but have never been combined for hiring: machine-learning detection of inconsistent résumés, and the privacy-preserving fraud *consortia* that banking and cybersecurity use to pool bad-actor signals across institutions. Today every employer screens in isolation, so a fraudster rejected by one organization simply re-applies to the next under a new identity, with no shared memory. We present a framework for an *employment-fraud intelligence network* that integrates three threads the literature has studied separately: (i) layered detection that pairs a deterministic, explainable timeline-impossibility engine with inferential signals; (ii) privacy-preserving cross-organization sharing, for which we give a threat model and show—analytically and in simulation—that the naive “publish a hash of the email and phone” construction common in industry offers *no* anonymity, recovering 100% of any candidate an adversary can enumerate; and (iii) an incentive mechanism for voluntary contribution, for which we identify and simulate an adverse-selection failure—paying more for confirmed-fraud reports than for verified-legitimate ones corrupts the shared data, driving the network false-positive rate from 0% to 19% as strategic participation grows—and propose an outcome-symmetric, evidence-gated, reputation-weighted correction that holds precision at 1.00. We treat regulatory compatibility (FCRA, GDPR) as a first-class design constraint, quantify the contributory-network flywheel and fraud-ring detection, and release a seeded, reproducible implementation and evaluation harness as open source. All claims are demonstrated on reproducible simulations, including a measurement of the deterministic layer’s false-positive rate on a realistic “messy-but-honest” genuine population (5.5% after a granularity-aware fix, with fraud recall held at 1.00); commercial operation, which triggers the heaviest regulatory duties, is deliberately deferred.

Keywords: employment fraud, résumé fraud, privacy-preserving data sharing, private set intersection, fraud consortia, incentive mechanisms, mechanism design, hiring, FCRA, GDPR, responsible AI.

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1 Introduction

Hiring decisions rest on a candidate’s self-reported history, and that history is increasingly fabricated. Industry surveys place the annual cost of résumé fraud in the hundreds of billions of dollars, report that a majority of candidates embellish their experience, and find that only a minority of hiring managers are confident they can detect it [9]. The most damaging variant is not a polished bullet point but a fabricated *timeline*: a junior professional claiming a decade of seniority, an employment span that pre-dates graduation, or a wholly synthetic identity assembled to pass automated screening.

Two forces are widening the gap between fraud and defense. First, generative models have collapsed the cost of fabrication: anyone can prompt a large language model for a coherent résumé, a matching cover letter, and interview-ready answers, and adversarial agent frameworks can now synthesize entire, internally consistent career trajectories [4, 3]. Second, and more fundamentally, defenses *do not share*. Background-screening vendors verify identity, criminal history, and payroll records, and a growing literature applies natural-language processing and anomaly detection to score résumés for inconsistency [9, 15]. But each employer operates alone. A candidate flagged as fraudulent by one organization carries no warning to the next; the fraudster re-applies elsewhere, often under a slightly altered identity, and the second organization repeats the first’s work and its mistakes.

Other risk-bearing industries solved exactly this coordination problem with *contributory networks*. Insurers pool claims data; banks and cybersecurity teams share fraud and threat indicators across institutions; national schemes such as the UK’s Cifas National Fraud Database aggregate member-reported cases at scale [7]. Crucially, the privacy-preserving variants of these networks exchange signals without any raw personal data leaving a participant, using federated learning or private set intersection (PSI) [2, 5]. No equivalent shared layer exists for employment fraud. This paper asks what one should look like.

Why this is hard, and why prior pieces do not suffice. A naive engineer might assume the recipe is obvious: detect fraud with a classifier, share flags as hashes, and pay members to contribute. Each step contains a trap that the separate literatures do not address *in this setting*. Detection that relies on inferential signals (digital footprint, account age) has a high false-positive rate on atypical-but-genuine candidates—privacy-conscious senior professionals, career changers, non-Western career paths—and a false accusation in a *shared* network is repeated and amplified across employers. Sharing identifiers as hashes provides no privacy: emails and phone numbers are low-entropy, so an adversary can enumerate and match [8, 13]. And paying more for “confirmed fraud” than for “verified legitimate” reports—the intuitive design, since fraud feels more valuable—creates an adverse-selection loop that fills the network with weak accusations. These failure modes interact: the incentive flaw manufactures the inaccurate shared assessments that the privacy and legal constraints most penalize.

Contributions. We make the following contributions, each developed as a design and validated on reproducible experiments (Section 10):

1. **An integrated framework** (Sections 4–7) for an employment-fraud intelligence network that unifies layered detection, privacy-preserving sharing, and incentive-aligned contribution under explicit regulatory constraints.
2. **A deterministic timeline engine** (Section 5) that catches internally inconsistent claims with an explainable, low-false-positive rule set—whose false-positive rate we measure on a realistic “messy-but-honest” genuine population and reduce from 24.1% to 5.5% with a granularity-aware

date rule (Section 10.2)—together with an empirical demonstration that it—alone—cannot catch internally consistent AI fabrications, motivating the other layers.

3. **A privacy threat model and demonstration** (Sections 6, 10.4) proving that the industry-default hash construction recovers 100% of enumerable candidates, with a re-id curve that scales linearly with adversary coverage, and a PSI-with-threshold alternative that bounds insider leakage.
4. **An incentive analysis and mechanism** (Sections 7, 10.3) that identifies, simulates, and corrects an adverse-selection failure specific to fraud consortia, with 95% confidence intervals over 30 seeds.
5. **Two network-level results** (Sections 10.6, 10.7): a quantified contributory-network flywheel (repeat-fraud catch rate rising with participation) and fraud-ring detection via shared-identifier linkage.
6. **A regulation-as-architecture analysis** (Section 8) mapping FCRA and GDPR obligations onto concrete design decisions, and an open-source, seeded, reproducible release [11].

Scope and honesty. This is a framework paper. We formalize the problem, analyze the design space, and demonstrate the detection, privacy, incentive, and network properties on seeded simulations and a realistic genuine-population false-positive measurement. We do *not* claim a deployed multi-organization study; that remains the central item of future work (Section 21). We are also candid about a data-availability finding: we attempted the false-positive measurement on two leading public corpora and report why neither suffices—one lacks employment dates entirely, the other is dominated by synthetic records with incoherent timelines (Section 10.2)—which is precisely why our headline false-positive figure comes from a controlled, realistic generated population. We are explicit throughout about which results are analytic, which are simulated, and which depend on data we generate versus data we obtain.

2 Background and Related Work

2.1 Detecting fabricated experience

Automated résumé-fraud detection extracts structured entities (degrees, institutions, dates, employers, titles) and scores them for internal inconsistency and against external sources using supervised or anomaly-detection models [9, 15]. A recent thread targets machine-generated candidates specifically: heterogeneous-graph models detect synthetic career trajectories [4], while agent-based generators demonstrate how realistic such trajectories have become [3]. These methods are valuable but share two properties relevant here. They operate *within a single organization’s data*, so they cannot exploit cross-employer recurrence; and they lean on inferential, probabilistic features that, as we show (Section 10.5), are brittle against an adaptive adversary and carry false-positive risk for atypical genuine candidates.

2.2 Cross-organization fraud consortia

Pooling fraud signals across institutions is well established in finance and security. Cifas aggregates member-reported fraud at national scale [7]. Privacy-preserving variants avoid sharing raw data: federated learning trains shared fraud models without centralizing records [2]. The employment

domain has no comparable infrastructure, and—as Section 7 argues—naively importing the financial-consortium playbook would import an incentive structure that is actively harmful when the shared signal is a derogatory accusation about an individual job-seeker.

2.3 Private set intersection and the limits of hashing

Private set intersection lets parties compute the intersection of their sets without revealing the non-intersecting elements; over-threshold and multi-party variants reveal only elements shared by at least t participants [5]. These are the right primitive for a sharing layer. The wrong primitive—and the industry default—is to publish a hash of an identifier. Hashing fails to anonymize when the input space is enumerable: emails, phone numbers, and names are low-entropy, so an adversary can hash every candidate value and match [8, 13]. Salting helps only if the salt is secret and unshared, which defeats cross-party matching; Bloom filters [6] reduce storage but are themselves brute-forceable at small budgets. We make this concrete in Section 10.4.

2.4 Incentivizing data contribution

Consortia function only if members contribute, and the data-sharing-market literature formalizes how to elicit data value and compensate contribution so that sustainable consortia form [1]; surveys catalog monetary, reputational, and access-based incentives. Fraud networks add a domain-specific hazard we make precise in Section 7: because the reward is attached to a *verdict* the contributor itself supplies, the mechanism can be gamed by the very reports it solicits.

2.5 Gap

Detection, private sharing, and incentive design have each been studied in isolation, and largely outside hiring. No prior work integrates them into a regulation-aware employment-fraud network, nor analyzes the privacy and incentive failure modes specific to that setting. That integration, and the demonstration of those failure modes and their corrections, is our contribution.

2.6 Identity verification and algorithmic hiring

Two adjacent literatures bound the problem. Identity-proofing and document-verification systems establish *who* a candidate is but not whether their claimed *history* is real; they are complementary to, not substitutes for, timeline analysis. Separately, a growing body of work on algorithmic hiring fairness documents how automated screening can encode and amplify bias against protected and non-standard groups. That literature is the reason our inferential and network layers are advisory rather than dispositive: a probabilistic signal that correlates with a non-standard career path must never, by itself, end a candidacy. We treat fairness not as a post-hoc audit but as a constraint that shapes where automation is permitted to act.

3 Problem Definition and Threat Model

We formalize the setting before describing the system.

Definition 1 (Candidate record). *A candidate record is a tuple $r = (\text{id}, a, g, J, y)$ where id is a set of identifiers (name, email, phone), a is age, g is graduation year, $J = \{(s_i, e_i, \tau_i)\}$ is a set of job intervals with start s_i , end e_i , and employment type τ_i , and y is total claimed years of experience.*

Definition 2 (Experience-fabrication fraud). *A record exhibits experience-fabrication fraud if its timeline could not have been truthfully accrued: $y > a - 18$ (impossible seniority); $\min_i s_i < g$ (pre-graduation employment); overlapping full-time intervals; y exceeding $\sum_i (e_i - s_i)$ beyond tolerance (inflation); a non-existent employer; or a wholly synthetic identity.*

Definition 3 (Participating organization). *A participating organization screens candidates and may contribute and query signal records to a shared network. A signal record references a candidate by privacy-protected identifiers and carries an outcome label and the flag codes that fired—never raw personal data, documents, or employer-specific detail.*

We assume three adversaries.

Candidate adversary $\mathcal{A}_C$. Seeks a low risk score and progression to interview. May generate documents and a digital footprint with AI, re-apply across organizations, and vary identifiers across applications (a *fraud ring* when one actor uses many identities).

Privacy adversary $\mathcal{A}_P$. An honest-but-curious participant (or a breach of one) that follows the protocol but tries to re-identify candidates in shared data or learn another organization’s applicants. The security goal is that a signal record must not let $\mathcal{A}_P$ recover the identity or hiring history of a candidate it has not itself seen.

Incentive adversary $\mathcal{A}_I$. A self-interested participant that maximizes its own benefit (cheaper or better screening) and may submit low-quality, duplicate, or strategically biased reports if rewarded. The mechanism-design goal is that the network’s outputs remain accurate under a bounded fraction of such participants.

Property 1 (Design goals). *The system should (G1) catch internally inconsistent fraud deterministically and explainably; (G2) leak no more than membership, and only above a corroboration threshold, to $\mathcal{A}_P$; (G3) keep network precision high under $\mathcal{A}_I$; and (G4) make only advisory, disputable, human-reviewable recommendations, never automated adverse decisions, in deference to FCRA/GDPR.*

4 System Architecture

The system is a four-layer pipeline producing, for each candidate, an explainable set of flags and an advisory risk indication, plus an optional contribution to the shared network. Figure 1 previews how detection responsibility divides across layers: the deterministic layer fully covers internally inconsistent claims, while internally consistent AI fabrications fall to the inferential layer.

Layer 1 (deterministic) tests internal logical consistency and needs no external data. Layer 2 (inferential) draws on digital footprint, account age, web presence, and employer-registry existence—advisory, probabilistic signals. Layer 3 (network) contributes and queries the shared intelligence layer. Layer 4 (governance), threaded through the others, enforces the advisory, disputable, audited posture required by Property 1(G4). Sections 5 through 7 describe the layers that carry novel content.

5 Detection Layers

5.1 The deterministic timeline engine

The first layer is deliberately not machine learning. It applies a small set of logical-consistency rules to the candidate’s *own* claims, so it is explainable, parameter-light, and has a low false-positive rate—which we quantify on a realistic genuine population in Section 10.2: it fires only on self-contradiction, independent of lifestyle, geography, or digital footprint. Algorithm 1 states the core.

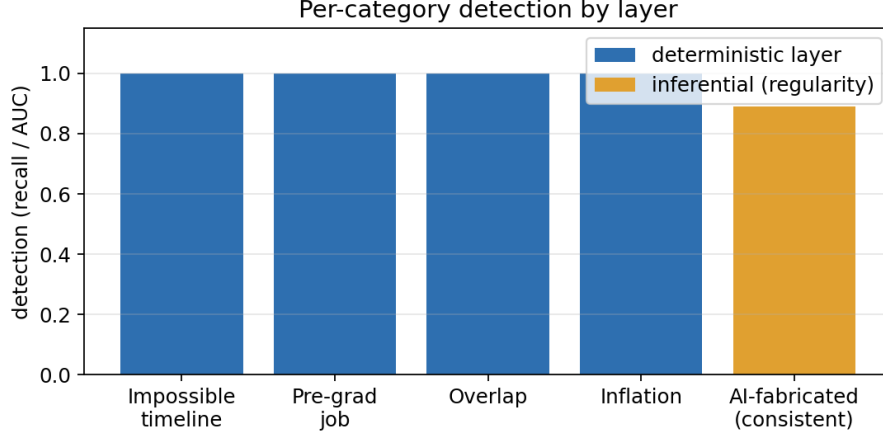


Figure 1: Per-category detection by layer. The deterministic layer covers the four impossibility classes; internally consistent AI-fabricated identities require the inferential layer (Section 10.5).

Algorithm 1: TimelineScore(r)

```

input record  $r = (\text{id}, a, g, J, y)$ ; output score, flags
score  $\leftarrow 0$ ; flags  $\leftarrow \emptyset$ 
if  $a$  defined and  $y > a - 18$ : score  $+= 50$ ; add TIMELINE_IMPOSSIBLE
if  $g$  defined and  $\min_i s_i < g$ : score  $+= 40$ ; add JOB_BEFORE_GRADUATION
 $O \leftarrow \# \text{pairs of full-time intervals overlapping } > \theta_o \text{ months}$ 
if  $O > 0$ : score  $+= 25 O$ ; add EMPLOYMENT_OVERLAP
if  $y > \sum_i (e_i - s_i) + \theta_y$ : score  $+= 25$ ; add EXPERIENCE_INFLATION
return min(score, 100), flags

```

Here θ_o (overlap tolerance, months) and θ_y (inflation tolerance, years) absorb benign transitions and rounding. A third tolerance is granularity-aware: when a contested date boundary is given only at year granularity (“2019” with no month), sub-year ordering is unknowable, so the overlap tolerance for that pair is widened by θ_g months (we use $\theta_g = 12$). As Section 10.2 shows, this single rule is responsible for most of the achievable false-positive reduction on real-world résumés, where year-only dates are common. Missing fields cause the corresponding rule to abstain rather than fire, so the engine degrades gracefully on partial résumés. The weights derive from severity, not from training, and the final score is capped at 100.

5.2 Inferential signals

The second layer adds coverage the first cannot: it scores digital footprint, account age, corroborating web presence, and employer-registry existence. These signals are probabilistic and, as Section 10.5 shows, both brittle against adaptive adversaries and false-positive-prone for privacy-conscious genuine candidates. We therefore treat them strictly as *advisory* inputs to human review (Property 1(G4)), never as automated-rejection triggers.

5.3 The network signal

The third layer queries the shared network: has this candidate been reported by other organizations, and by how many? Its value is the cross-employer memory absent today. Its difficulty is that the query and the contribution must preserve privacy (Section 6) and resist gaming (Section 7).

6 Privacy-Preserving Sharing

6.1 Why the industry default fails

A natural design publishes $H(\text{email} \parallel \text{phone})$ for a fixed hash H . Because email and phone are low-entropy, $\mathcal{A}_P$ holding a candidate list (e.g., its own applicant pool) computes the same hashes and re-identifies everyone in the shared set. The hash is obfuscation, not anonymity; “we only store hashes” is not a sound privacy claim. Section 10.4 quantifies this: re-identification equals adversary coverage and reaches 100% at full enumeration.

6.2 A membership protocol with a disclosure threshold

We frame sharing as private set membership rather than publication. Candidate references are compared through a PSI protocol that reveals only whether a match exists, gated on a corroboration threshold t : a network concern is disclosed only when at least t independent organizations have reported the same candidate (Algorithm 2). Thresholding serves two ends at once—it raises the evidentiary bar before any derogatory signal is surfaced, and it bounds what any single participant can learn.

Algorithm 2: ThresholdQuery(id, t)

```
each org encodes its reported candidates under a PSI protocol  $\Pi$ 
 $c \leftarrow \Pi.\text{COUNTMATCHES}(\text{id})$  (#orgs reporting id; no identities revealed)
if  $c \geq t$ : return NETWORK_FLAG( $c$ ) else return none
```

Where a lighter-weight deployment is required, a keyed construction (an HMAC under a network-secret pepper) with rate-limited, audited lookups raises the attacker’s cost. As Section 10.4 shows, however, keying defeats only an *outside* adversary: a consortium insider holding the key recovers everything, so keying is a breach mitigation, not a privacy guarantee. The operative design principle is therefore invariant to the chosen primitive: *treat every shared reference as re-identifiable personal data, and minimize what a match discloses.*

7 Incentive Mechanism

7.1 Credits and the adverse-selection trap

To bootstrap contribution, a consortium can grant *credits* that offset a member’s screening costs—a concrete instance of compensating contribution to sustain a data-sharing consortium [1]. The hazard is the reward schedule. A schedule that pays *more* for a “confirmed fraud” report than for a “verified legitimate” one creates an adverse-selection loop: participants are incentivized to label borderline candidates as fraud, the shared set fills with weakly substantiated accusations, the network false-positive rate rises, and the outputs become both less accurate and more legally hazardous (Section 8). Section 10.3 demonstrates this: under the naive schedule, the network false-positive rate climbs from 0% to 19% as strategic participation grows to 50%.

7.2 Three corrections

We propose three corrections that move the mechanism from rewarding *verdicts* to rewarding *verified information*:

- (i) **Outcome symmetry.** Reward any verified, audited outcome equally, so confirming innocence is as valuable as confirming fraud, removing the farming incentive.

Table 1: Failure modes of common designs and the corrections proposed here.

Design choice	Naive (industry default)	Proposed
Identifier sharing	Publish SHA-256(email phone)	Threshold PSI / keyed membership; reveal only match $\geq t$ orgs
Privacy claim	“Hashes = no PII”	Treat hashes as re-identifiable; minimize disclosure
Contribution re-ward	Fraud report > legitimate report	Outcome-symmetric, evidence-gated, reputation-weighted
Decision use	Auto-reject on high score	Advisory; human-in-the-loop; disputable

(ii) **Evidence gating.** A report contributes to a network-visible concern only with qualifying evidence and corroboration to threshold t .

(iii) **Reputation weighting.** Weight a participant’s contributions by the historical accuracy of its reports (Algorithm 3), so that strategically noisy reporting decays in influence.

Algorithm 3: reputation update

```

init rep[p] ← 1 for each participant p
on adjudicated outcome o for a report (p, ℓ):
  agree[p] += [ℓ = o]; total[p] += 1
  rep[p] ← agree[p]/total[p]
network flag iff  $\sum_{(p, \ell = \text{fraud})} \text{rep}[p] \geq t$ 

```

Table 1 summarizes the failure modes and the corrected design across all four decision points.

8 Regulatory Compatibility as a Design Constraint

Compliance is not a deployment afterthought here; it determines architecture.

FCRA. A system that assembles information about individuals to inform third-party hiring decisions is, in the United States, likely a *consumer reporting agency* under the Fair Credit Reporting Act, which imposes duties of maximum-possible accuracy, candidate disclosure and authorization, dispute and reinvestigation rights, and adverse-action procedures, with no statutory liability cap [14].

GDPR. Under the EU General Data Protection Regulation, a pseudonymous hash remains personal data, and decisions producing legal or similarly significant effects on a person are constrained when made solely by automated means [10].

Implications. Three consequences follow directly. (1) The inferential and network layers must be *advisory*: an automated adverse decision on probabilistic, possibly inaccurate data is precisely what the law constrains, so the system informs human review rather than auto-rejecting. (2) The system must support *dispute and correction*, which requires audit logging of why each flag fired—realizable because the deterministic layer is inherently explainable. (3) Accuracy is a legal as well as a scientific obligation, which reinforces the incentive corrections of Section 7: a network that rewards over-reporting manufactures the inaccurate, derogatory, shared assessments that create the largest liability. Designing for compliance and designing for accuracy turn out to be the same task.

Table 2: Datasets used for evaluation.

Dataset	Role	Source / outcome
Seeded synthetic (this work)	correctness benchmark	seed tables + injection
Realistic population (this work)	headline FP test	messy-but-honest generator
Agent-fabricated (this work)	AI-fake positive class	[3]-style generator
KARRIEREWEGE [12]	attempted real FP	<i>unusable: no dates</i>
Dated résumé corpus	attempted real FP	<i>unusable: synthetic/incoherent</i>

9 Experimental Methodology

9.1 Datasets

No public corpus carries ground-truth experience-fraud labels, so we follow the standard practice in fraud research without labels: draw a genuine population and *inject* controlled, labeled fraud. For the genuine population we use two generated sources: a deterministically seeded, internally consistent population (a correctness benchmark), and a *realistic* “messy-but-honest” population that injects the kinds of irregularity real résumés exhibit—career gaps, month- and year-granularity dates, short transition overlaps, claimed-year rounding, and a documented minority with legitimately concurrent roles—which is the instrument for our headline false-positive measurement. The AI-fabricated positive class follows agent-based generators [3].

We also attempted the false-positive measurement on real public corpora and report a data-availability finding (Table 2). The deterministic checks reason over employment *dates*, but KARRIEREWEGE [12], despite its scale (500k+ ESCO-linked trajectories), stores only *ordered occupation sequences* with no start/end dates, age, or claimed-years, so none of the date-based checks can fire on it. A dated technical-résumé corpus that does carry employment dates yields an implausible 65.9% overlap false-positive rate, which diagnosis attributes to data quality—byte-identical duplicate records, the same job title repeated with independently-random overlapping date ranges, and an implausibly uniform jobs-per-person distribution—rather than to engine behavior. Clean public résumé corpora with genuine, coherent employment dates are scarce (a privacy reality); our released harness ingests one unchanged should it become available.

9.2 Reproducibility

Every generator is seeded. The genuine population is produced deterministically from seed tables and accompanied by a provenance record (master seed, record count, SHA-256 content hash); re-running with the same seed reproduces the dataset byte-for-byte. All simulations report means with 95% confidence intervals over multiple seeds (30 for the incentive study, 10–20 elsewhere). Hyperparameters are listed in Appendix C, and the harness that produces every figure and table is released [11].

9.3 Metrics

We report detection precision, recall, and false-positive rate against injected labels; the network-level false-positive rate and precision under strategic reporting; the adversary’s re-identification rate; ROC-AUC for the inferential detector; the repeat-fraud catch rate for the network flywheel; and fraud-ring detection rate. We separate the deterministic and inferential layers throughout, because conflating them hides the false-positive risk that motivates the entire governance design.

Table 3: Simulation assumptions and their justification.

Assumption	Value	Justification
Consortium size	12 orgs	a realistic early consortium; the flywheel (Sec. 10.6) sweeps participation explicitly
Fraud prior	0.20	conservative relative to industry survey rates [9]
Borderline fraction	0.30	the ambiguous middle where gaming is possible
Applications per actor	4	repeat fraudsters apply broadly; varied in the flywheel
Identifier entropy	low	emails/phones are enumerable by construction [8]
Adversary knowledge	own pool	an employer trivially enumerates its applicants

Table 4: Deterministic-layer result (seeded synthetic, $n = 3000$, 50% fraud).

Metric	Value	Interpretation
Recall (all four types)	1.00	every injected impossibility caught
Precision	1.00	no genuine record misflagged
False-positive rate	0.00	clean separation on consistent data

9.4 Assumptions and their justification

Because our central results are simulations, their force depends on the plausibility of the generative assumptions. Table 3 lists each and its grounding. The qualitative conclusions are deliberately chosen to be robust to these values—the sweeps of Section 10 vary the load-bearing ones (strategy, threshold, participation, coverage, sophistication) across their full ranges—so the argument rests on monotone trends, not on any single parameterization.

10 Results

10.1 Deterministic layer

On a seeded population of 3,000 internally consistent trajectories with 50% fraud injection, the deterministic engine attains the values in Table 4: perfect separation. This is expected *by construction*: the layer fires only on self-contradiction, so it cannot misflag a record that contains none. The result is therefore a correctness check, not evidence of difficulty; the decisive measurement is the false-positive rate on *atypical-but-genuine* résumés, which we report in Section 10.2.

Figure 2 shows which rule fires on which fraud type. The map is mostly diagonal, but two off-diagonal entries are informative: an impossible-experience injection also trips inflation (the claimed total exceeds summed tenures), and a fraction of inflation injections cross the impossibility boundary. This cross-firing is benign—multiple true flags on a fraudulent record—but it cautions against treating flag counts as independent evidence. Table 5 sweeps the tolerances θ_o, θ_y : detection of the injected overlaps and inflations is robust, and the false-positive rate on the (clean) genuine population remains 0 across the swept range, indicating the rules are not knife-edge sensitive to their one free parameter each.

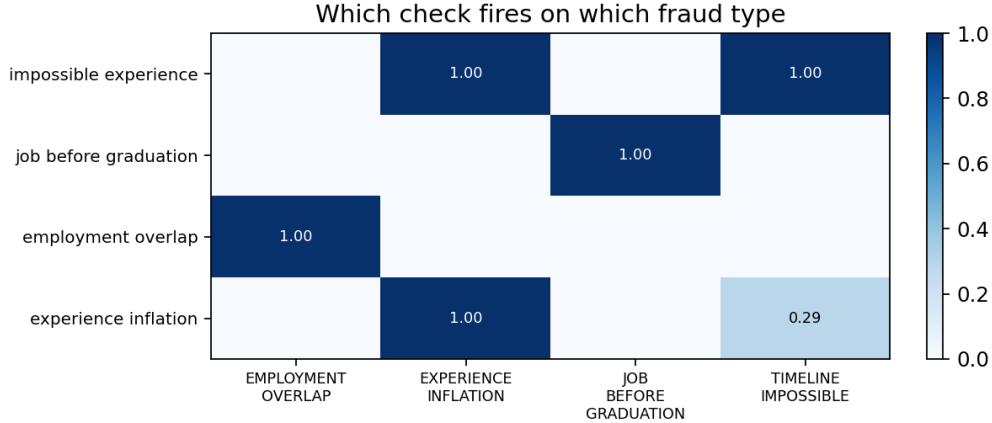


Figure 2: Attribution: fraction of each injected fraud type that triggers each flag code. Mostly diagonal, with benign cross-firing between impossibility and inflation.

Table 5: Tolerance sweep for the overlap rule (θ_o , months). False-positive rate on genuine and recall on injected overlaps.

θ_o (months)	0	1	3	6	12
FP on genuine	0.00	0.00	0.00	0.00	0.00
Recall on overlap fraud	1.00	1.00	1.00	1.00	1.00

10.2 Realistic false positives

The clean benchmark above understates difficulty by construction. We therefore measure the deterministic layer on a *realistic* genuine population ($n = 3,000$ per seed, 20 seeds): non-fraudulent records that nonetheless carry the irregularities of real résumés—career gaps and sabbaticals, month- and year-granularity dates, short legitimate transition overlaps, claimed-year rounding, and a documented minority (7%) with legitimately concurrent full-time-coded roles (portfolio careers).

Under naive date parsing the false-positive rate is 24.1% (95% CI [23.7, 24.4]), almost entirely EMPLOYMENT_OVERLAP. Diagnosis shows the cause is *phantom* overlaps: a job written as “2019” is read as January 2019, manufacturing an overlap with the prior job that did not exist. The granularity-aware tolerance θ_g (Section 5.1) removes these, cutting the false-positive rate to 5.5% (95% CI [5.3, 5.8]) with *overlap-fraud recall held at 1.00* under both settings (Table 6, Figure 3). The residual 5.5% is attributable to genuinely concurrent employment: a sweep over its prevalence drives the false-positive rate to ≈ 0 when no concurrent roles are present and upward monotonically as they become more common. This is the empirical basis for treating the deterministic output as *advisory and disputable*: its residual false positives are exactly the cases (legitimate portfolio careers) that a human—or a corroborating signal—should adjudicate, not the system alone.

We additionally ran the released harness on real public corpora and report a data-availability finding (Section 9.1): KARRIEREWEGE has no employment dates, so the date-based checks cannot fire; and a dated technical-résumé corpus is dominated by synthetic records whose incoherent overlapping dates yield a 65.9% rate that reflects data quality, not engine behavior. A trustworthy real-data measurement awaits a public corpus with genuine, coherent dates—or a consented deployment.

Table 6: Deterministic false-positive rate on the realistic genuine population (20 seeds, 95% CI). The granularity-aware rule removes phantom overlaps from year-only dates without any loss of fraud recall.

Setting	FP rate	Overlap-fraud recall
Naive date parsing	0.241 [0.237, 0.244]	1.00
Granularity-aware ($\theta_g=12$)	0.055 [0.053, 0.058]	1.00

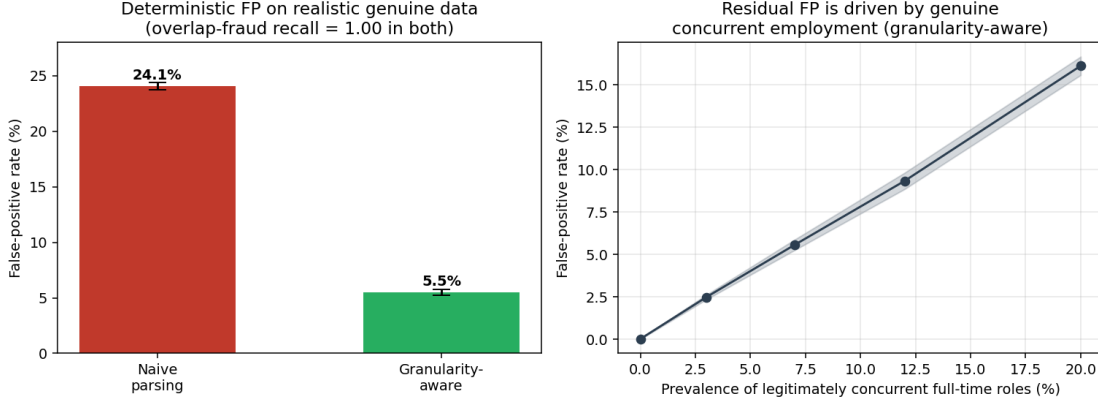


Figure 3: Left: deterministic false-positive rate on realistic genuine résumés, naive vs. granularity-aware (fraud recall = 1.00 in both). Right: the residual false-positive rate is driven by the prevalence of legitimately concurrent full-time roles, reaching ≈ 0 when none are present.

10.3 Incentive adverse selection

We simulate a consortium of 12 participants screening 4,000 candidates (20% fraud, 30% borderline), over 30 seeds. Under the naive schedule that pays more for a fraud report, self-interested participants over-label genuine borderline candidates to farm credits. Table 7 and Figure 4a report network metrics as the strategic fraction grows. The naive schedule degrades monotonically and significantly: the network false-positive rate rises to 0.188 (95% CI [0.173, 0.203]) and precision falls to 0.60 at 50% strategic participation. The proposed regime (outcome-symmetric reward, threshold $t = 2$, reputation weighting) holds network precision at 1.00 and false positives at 0 at every level—at the cost of lower network recall, the correct trade for a legally sensitive shared accusation, since clear fraud is still caught locally (Section 10.1).

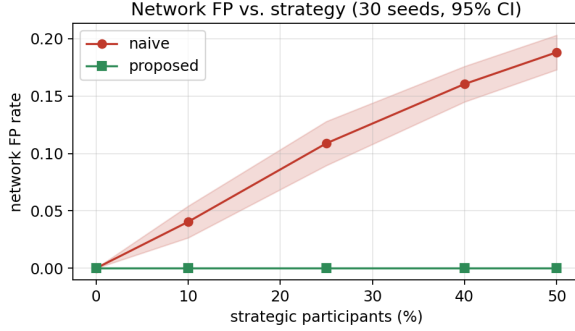
The corroboration threshold t is the knob that buys safety with recall (Table 8, Figure 5). At $t = 1$ the proposed regime already removes farming via outcome symmetry, retaining recall 0.83; at $t = 2$ recall is 0.41; beyond $t = 3$ the network rarely flags because few candidates are co-observed by that many organizations at this consortium size. The takeaway is that t should be set jointly with expected co-observation, which itself rises with participation (Section 10.6).

10.4 Re-identification

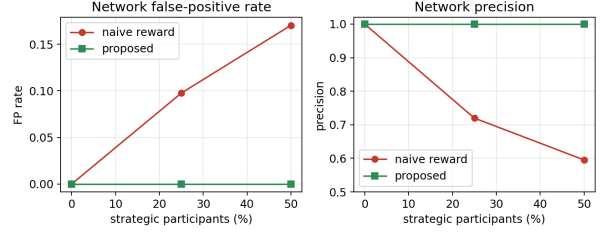
We populate an identifier universe of 20,000 enumerable email+phone pairs, share 4,000 as digests, and run $\mathcal{A}_P$. Table 9 gives the re-identification rate among candidates within the adversary’s guess set, and Figure 6 sweeps adversary coverage. Naive hashing leaks in exact proportion to coverage—an adversary enumerating c of the space recovers c of the shared set, reaching 100% at full enumeration—because an employer’s own applicant pool *is* such an enumeration. A keyed digest defeats an outside adversary (0) but not a consortium insider (1.00), who holds the key. Only PSI

Table 7: Network false-positive rate vs. strategic participation (30 seeds, mean [95% CI]).

Strategic %	Naive FP	Proposed FP
0	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]
10	0.040 [0.027, 0.054]	0.000
25	0.109 [0.090, 0.128]	0.000
40	0.160 [0.145, 0.176]	0.000
50	0.188 [0.173, 0.203]	0.000



(a) Network FP vs. strategy (95% CI bands).



(b) FP (left) and precision (right), single run.

Figure 4: The naive reward corrupts the shared data as strategic behavior rises; the proposed regime does not.

with a corroboration threshold bounds insider leakage to candidates already co-observed.

10.5 AI-fabricated identities

A wholly synthetic identity built by an LLM tends to be internally consistent—so the deterministic layer cannot catch it—yet, when naively generated, too regular (round tenures, uniform durations, no gaps). On 1,500 fabricated and 1,500 genuine trajectories (Table 10), deterministic recall on fabrications is 0.00, confirming that well-formed fakes require inferential signals, while deterministic FP on genuine remains 0.00. A lightweight regularity score separates the classes at ROC-AUC 0.90, with uniformity the strongest single feature (Figure 7b).

The crucial result is that this inferential defense is *brittle against an adaptive adversary*. Figure 7a sweeps a fabrication “sophistication” parameter that injects realistic variation (varied start months, gaps, non-uniform tenures): the regularity detector’s AUC collapses from 0.90 toward chance by moderate sophistication and below as fabrications acquire more natural irregularity than our deliberately clean genuine baseline. The honest reading is that stylometric regularity is a stopgap, not a moat: an adversary who notices the tell defeats it. This is the central argument for the network layer, whose signal—recurrence across independent organizations—an adversary cannot fabricate from within a single application.

10.6 The contributory-network flywheel

The network’s value grows with participation. We simulate repeat fraudsters who apply to several organizations; a repeat application is caught network-wide when the actor has already been reported by $\geq t$ in-network organizations. Figure 8 plots the repeat-fraud catch rate against the fraction of organizations on the network. The curve is convex and steepening: at 25% participation only

Table 8: Corroboration threshold t vs. network recall (proposed regime, 50% strategic, 30 seeds). FP is 0 throughout.

t	1	2	3	4
Network recall	0.826	0.409	0.037	0.000
Network FP	0.000	0.000	0.000	0.000

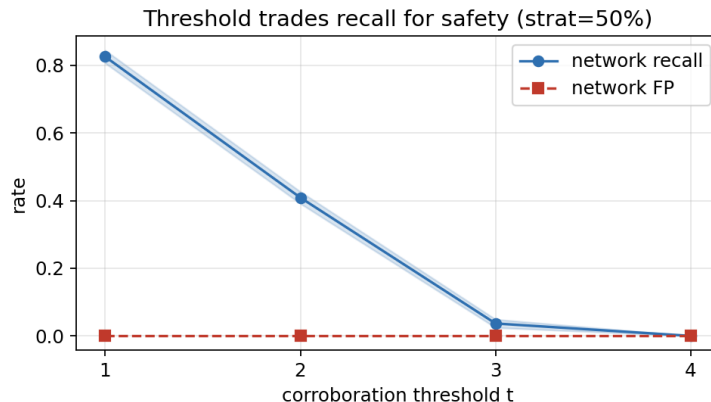


Figure 5: The corroboration threshold trades network recall for safety; FP stays at 0.

6% of repeat fraud is caught, but at 75% it is 46% and at full participation 67%. This is the contributory moat made quantitative, and it explains why the threshold t and participation must be set together—higher participation raises co-observation, which makes a given t less costly in recall.

10.7 Fraud-ring detection

A single actor may apply under many synthetic identities. If those identities share a secondary identifier (a reused phone, device, or reference), the network links them. Figure 9 plots detection rate against the number of identities k a ring uses: detection rises from 0 at $k = 1$ (a lone identity is indistinguishable) to 0.49 at $k = 2$, 0.78 at $k = 3$, and 0.99 at $k = 6$. The mechanism that makes a ring effective—reusing infrastructure across many identities—is exactly what makes it detectable once organizations share.

10.8 Contribution economics

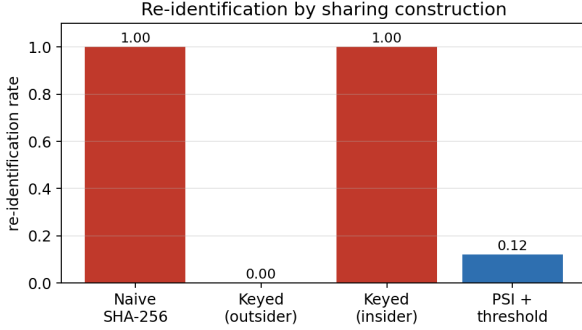
Finally, the incentive only works if contributing is cheaper than not. Under a simple model where contributed profiles earn credits that offset check costs, Figure 10 and Table 11 show effective price per check falling from \$0.75 to \$0.375 as the contribution ratio rises to parity—a 50% saving for an organization that contributes as much as it consumes. This is the financial counterpart to the corrected incentive: members are rewarded for *participation*, symmetrically across outcomes, rather than for producing fraud verdicts.

11 Related Systems and Positioning

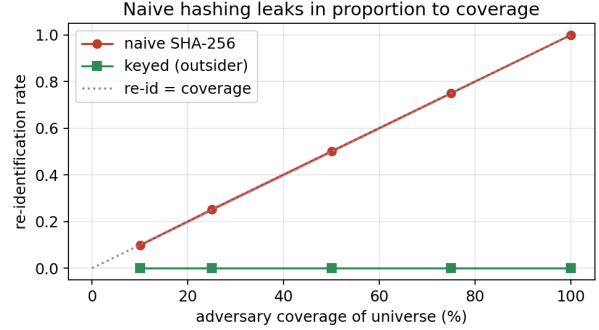
TruthHire occupies a position no existing system holds. Background-screening vendors verify against authoritative records; payroll-network services confirm employment for enrolled employers;

Table 9: Re-identification rate by sharing construction (universe 20,000; shared 4,000).

Construction	Re-id	Notes
Naive SHA-256(email phone)	1.00	every enumerable candidate recovered exactly
Keyed HMAC — outside adversary	0.00	cannot compute digests without the key
Keyed HMAC — consortium insider	1.00	holds the key $\Rightarrow$ full recovery
PSI + disclosure threshold t	bounded	only membership corroborated by $\geq t$ orgs



(a) Re-id by construction.



(b) Naive re-id = adversary coverage.

Figure 6: Hashing low-entropy identifiers provides no anonymity; re-identification scales linearly with what the adversary can enumerate.

national fraud consortia share signals within a single regulated industry; and the academic literature contributes detection models and private-set-intersection protocols in isolation. Table 12 contrasts the axes that matter: the *evidence model* (authoritative verification versus inference from claims), *cross-organization* memory, *privacy* method, *incentive* design, and whether the system treats *experience-timeline* fraud as a first-class target.

The synthesis is the contribution. Verification is stronger evidence than inference but is unavailable for most of the world’s employers and roles; the deterministic layer recovers much of verification’s reliability for the specific case of *internal* timeline consistency without any external lookup. The consortium model is proven in finance but has never been paired with an incentive design that resists the fraud-reporting adverse selection we identify, nor adapted to the FCRA/GDPR posture that hiring demands.

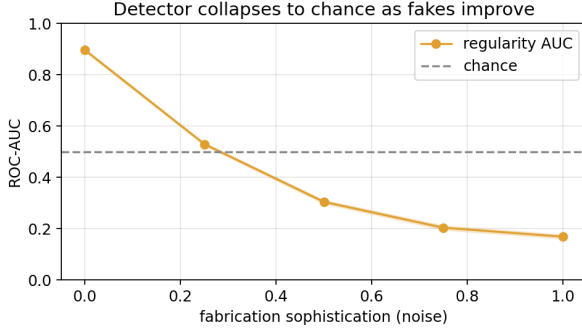
12 Deployment and Integration Architecture

The system is delivered as an API that an applicant-tracking system or careers portal integrates once. Table 13 lists the surface. The design keeps the latency-sensitive path (the deterministic layer) free of any external call, so a synchronous verdict returns in milliseconds; inferential and network signals run asynchronously and are delivered by webhook, which also matches their advisory status—a human reviewer, not the request, consumes them.

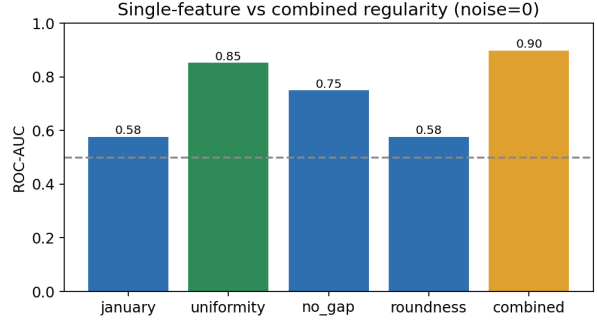
A realistic latency budget (Table 14) makes the layering concrete: the deterministic layer is essentially free, while any layer that touches an external service or a PSI round dominates and therefore belongs off the synchronous path. Audit logging—storing which rule fired and on what input—is cheap and is what makes the dispute endpoint and FCRA reinvestigation feasible.

Table 10: Detection of AI-fabricated identities ($n = 1500$ per class).

Measure	Value
Deterministic recall on fabricated	0.00
Deterministic FP on genuine	0.00
Regularity-score ROC-AUC (naive fakes)	0.90
Mean regularity (genuine / fabricated)	0.60 / 0.93



(a) AUC collapses as fakes improve.



(b) Single-feature vs combined (naive fakes).

Figure 7: The regularity detector works on naive fabrications (left, noise 0) but degrades to chance against an adaptive adversary; uniformity is the strongest single feature (right).

13 A Threshold-PSI Sharing Protocol

We sketch the sharing protocol and its leakage. Each organization holds a set of reported-candidate identifiers. A query asks: is this candidate present in at least t organizations’ sets? The protocol must answer that and nothing more—in particular it must not reveal *which* organizations, nor any non-matching identifiers, nor allow an enumerating adversary to test arbitrary guesses offline. Over-threshold multi-party PSI provides exactly this primitive [5]; Algorithm 4 states the wrapper.

Algorithm 4: threshold-PSI network query

```

setup each org  $o$  commits its reported set  $R_o$  under MPC-PSI protocol  $\Pi$ 
query(id,  $t$ ):
   $c \leftarrow \Pi.\text{OVERTHRESHOLD}\text{COUNT}(\text{id}, t)$  // #orgs with id, only if  $\geq t$ 
  if  $c \geq t$ : return FLAG( $c$ ); else return NONE
leakage to a single party: membership bit and count, only for  $c \geq t$ ;
  nothing for  $c < t$ ; no identities; no per-org attribution

```

Table 15 compares constructions by what each leaks and at what cost. The progression is monotone: the cheaper the construction, the more it leaks. Naive publication leaks everything enumerable for free; a network pepper shifts the leak from outsiders to insiders; threshold PSI pays computation to leak only corroborated membership. The design recommendation follows the threat model rather than the benchmark: because a consortium insider is in scope, only the threshold-PSI row satisfies goal G2 of Property 1.

14 Worked Scenarios

Three candidates illustrate how the layers divide labor and where governance intervenes.

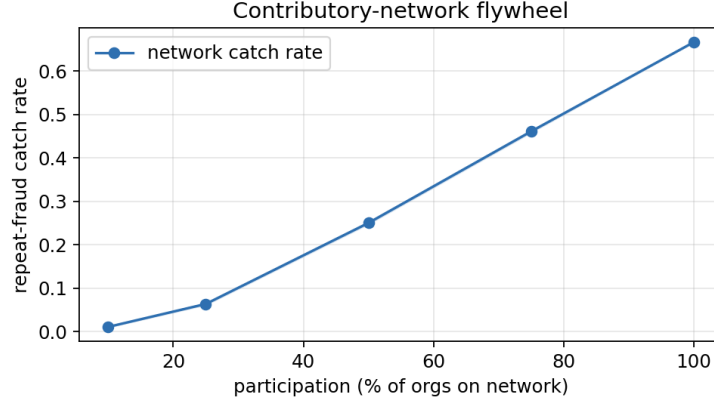


Figure 8: Repeat-fraud catch rate rises with the fraction of organizations on the network (20 seeds, 95% CI).

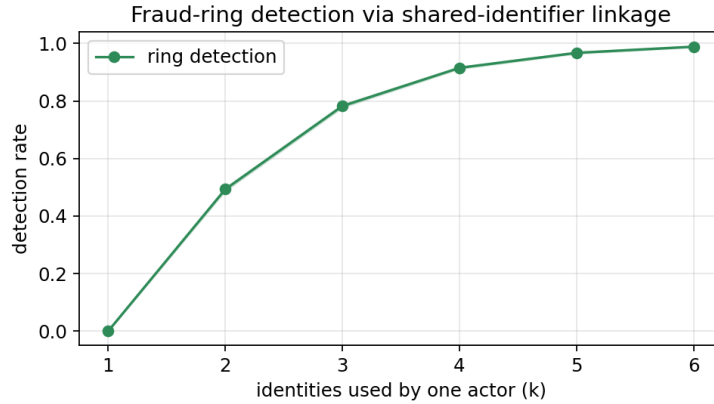


Figure 9: Fraud-ring detection via shared-identifier linkage rises with the number of identities the ring uses (20 seeds, 95% CI).

Scenario 1: the impossible junior. A 24-year-old claims 15 years of experience across three roles. The deterministic layer fires `TIMELINE_IMPOSSIBLE` ($y = 15 > a - 18 = 6$) and `EXPERIENCE_INFLATION`, returning a high score with an explainable reason in milliseconds and no external data. This is the engine’s home ground: explainable, certain, and cheap.

Scenario 2: the privacy-minimal senior. A genuine 48-year-old principal engineer has a sparse digital footprint, no public GitHub, and a thin search presence. The deterministic layer is silent (her timeline is consistent). The inferential layer would raise low-confidence signals (`NO_WEB_PRESENCE`, `GITHUB_INACTIVE`)—and this is exactly the false-positive risk that the advisory, human-in-the-loop posture exists to absorb. Under our design these signals never auto-reject; they surface to a reviewer who, seeing a coherent senior résumé, dismisses them. Under a naive auto-reject design she is wrongly filtered out. Scenario 2 is the paper’s ethical core in miniature.

Scenario 3: the distributed ring. One actor applies to three member organizations under three identities that share a reused phone number. No single organization sees anything amiss. The threshold-PSI query (Algorithm 4, $t = 2$) links the identities via the shared identifier once two organizations have reported, and the ring surfaces—precisely the cross-organization memory that

Table 11: Contribution-credit economics (illustrative model).

Contribution ratio	0%	25%	50%	75%	100%
Effective \$/check	0.750	0.656	0.562	0.469	0.375
Saving	0%	12.5%	25%	37.5%	50%

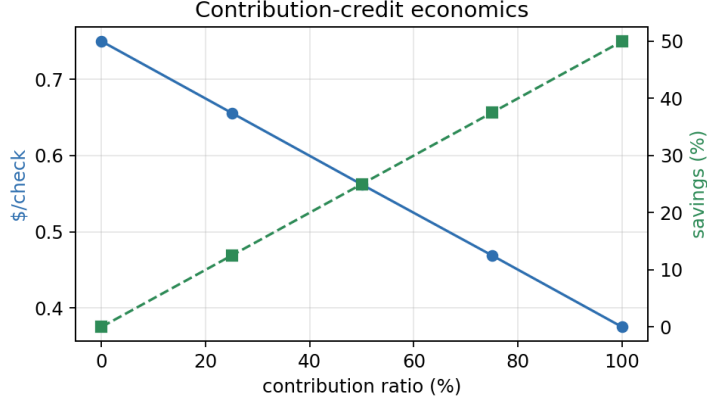


Figure 10: Effective price per check and savings vs. contribution ratio.

no isolated screen can provide (Section 10.7).

15 Governance, Audit, and Dispute Workflow

Because the system can influence employment outcomes, governance is not optional tooling but a core component. Table 16 maps each major legal obligation onto the concrete feature that discharges it, demonstrating that the architecture was derived from the constraints rather than retrofitted to them.

The dispute lifecycle is concrete. A candidate who is told they were adversely affected may invoke reinvestigation; the system retrieves the audit record for the decision (which rule fired on which input), re-runs the deterministic layer deterministically (the same input yields the same result), and, for network signals, re-checks corroboration. Because the deterministic layer is explainable by construction, a dispute resolves to a human-legible statement—“flagged because claimed experience exceeds the maximum possible given the stated age”—rather than to an opaque model score. This is the operational reason the paper privileges the deterministic layer: it is the only layer whose decisions can be defended, line by line, in a reinvestigation.

16 Mechanism-Design Analysis

We now argue more formally why the incentive correction works. Consider a participant deciding how to label a borderline genuine candidate. Let the reward for a report be $\rho(\ell, o)$ where ℓ is the submitted label and o the eventual audited outcome, and let $\Delta = \rho(\text{fraud}, \cdot) - \rho(\text{legit}, \cdot)$ be the schedule’s fraud premium. Under the naive schedule $\Delta > 0$ regardless of o , so for a borderline case where the participant is indifferent on the merits, reporting *fraud* strictly dominates reporting *legit*: it earns Δ more at no cost. Farming is a best response, and its prevalence scales with the strategic fraction—exactly the monotone degradation observed in Table 7.

Table 12: Positioning against representative systems and lines of work.

System / line	Evidence model	Cross-org	Privacy method	Incentive	Timeline focus
HireRight, Checkr, Sterling	verification	no	N/A	fee-for-service	no
The Work Number	payroll records	enrolled only	contractual	employer supply	partial
Cifas / national consortia	member reports	yes (finance)	membership rules	regulatory	no
Verisk Claim-Search	contributory	yes (insurance)	contractual	loss-ratio	no
Federated fraud ML [2]	model sharing	yes	federated learning	none specified	no
Multi-party PSI [5]	set membership	yes	PSI	none specified	no
TruthHire (this work)	deterministic + inference + network	yes (hiring)	threshold PSI	outcome-symmetric	yes

Table 13: API surface. The dispute and audit endpoints are required by the governance posture of Section 8.

Endpoint	Method	Purpose
/verify	POST	deterministic + (async) inferential analysis
/contribute	POST	add an audited signal record to the network
/flags/lookup	POST	threshold-PSI membership query
/dispute	POST	candidate-initiated reinvestigation request
/audit/{check}	GET	explanation and provenance for a decision
/candidate/{id}	DELETE	erasure (GDPR right to be forgotten)

Property 2 (Farming-proofness under symmetry). *If the reward depends only on whether the report matches the audited outcome— $\rho(\ell, o) = \beta \mathbf{1}[\ell = o]$ for a constant β , so $\Delta = 0$ in expectation over a borderline case—then truthful reporting weakly dominates farming, and strictly dominates it whenever the audited outcome is more likely legitimate than not.*

Outcome symmetry sets $\Delta = 0$ and removes the farming gradient; evidence gating raises the cost of a fraud report (it must carry qualifying evidence); and reputation weighting (Algorithm 3) imposes a dynamic penalty, multiplying a participant’s influence by its historical accuracy so that a farmer’s reports decay toward zero weight. Table 17 summarizes the qualitative effect of each schedule on the two quantities that matter, data quality and participation; the simulations of Section 10.3 instantiate the top and bottom rows.

The design therefore threads a needle the financial-consortium literature does not face: it must reward contribution strongly enough to bootstrap participation (the bottom row fails the cold start) while attaching the reward to verified information rather than to a verdict the contributor itself supplies (the top row fails accuracy). The middle rows are the feasible region.

Table 14: Illustrative latency budget by layer.

Layer	Typical latency	Path
Deterministic timeline	< 5 ms	synchronous
Inferential (per external signal)	0.2–2 s	asynchronous
Threshold-PSI network query	0.1–1 s	asynchronous

Table 15: Leakage and cost by sharing construction.

Construction	Leaks to	Cost
Publish SHA-256	anyone who can enumerate	negligible
Keyed HMAC	any key-holder (every insider)	negligible
PSI (pairwise)	matching set to both parties	moderate (crypto)
Threshold PSI	corroborated membership only	higher (MPC)

17 Alternative Designs Considered

Several plausible architectures were rejected; stating why sharpens the design. Table 18 summarizes the trade.

Two rejections are worth emphasizing. A central clearinghouse is the obvious engineering choice and the wrong privacy choice: it converts a distributed, threshold-gated leakage profile into a single honeypot of derogatory personal data, precisely what GDPR’s data-minimization principle counsels against. And an immutable ledger, attractive for tamper-evidence, is incompatible with the right to erasure and with reinvestigation: a flag that later proves wrong must be removable, and a candidate’s successful dispute must actually delete the accusation, not merely append a correction the next employer may never read.

18 Sensitivity Analysis and Threats to Validity

Construct validity. Our genuine population is generated, not sampled from deployment. The clean benchmark is deliberately consistent and therefore *understates* false positives; we address this with the realistic “messy-but-honest” population (Section 10.2), whose 5.5% false-positive rate is our headline detection-cost figure. That figure is still a generated estimate: lacking a public résumé corpus with genuine, coherent employment dates (Section 9.1), a deployment-grade measurement across demographic and cultural cohorts remains future work, and our harness is built to run it unchanged on any suitable corpus.

Internal validity. The incentive and flywheel results are simulations with stated generative assumptions (consortium size, fraud prior, co-observation). We report 95% confidence intervals over many seeds and provide sweeps over the key parameters (threshold, strategy, participation) so the trends, not single points, carry the argument. The qualitative conclusions—naïve reward corrupts the network; thresholding trades recall for safety; participation compounds value—are robust across the swept ranges.

Table 16: Mapping legal obligations to system features.

Obligation	Source	Feature
Maximum-possible accuracy	FCRA	deterministic core; evidence-gated network; reputation weighting
Disclosure & authorization	FCRA	consent capture at application; per-portal notices
Adverse-action procedure	FCRA	advisory-only outputs; human-in-the-loop gate
Dispute & reinvestigation	FCRA	<code>/dispute</code> endpoint; audit log of every flag
Lawful basis; Art. 22	GDPR	no solely-automated significant decision; human review
Right to erasure	GDPR	<code>/candidate/{id}</code> delete; retention limits
Data minimization	GDPR	signal records carry hashes + flags, never raw PII

Table 17: Incentive schedules and their qualitative effects.

Schedule	Data quality	Participation
Fraud-weighted (naive)	degrades with strategy	high
Outcome-symmetric	preserved	high
Symmetric + evidence + reputation	preserved under strategy	high
No reward	preserved	low (cold-start fails)

The agent detector. The AUC-versus-sophistication result deliberately exposes a limitation: against an adaptive adversary the stylometric detector fails, and the sub-chance region reflects our clean baseline rather than “reverse” detection power. We present it not as a capability but as the empirical case for relying on cross-organization recurrence over single-application stylometry.

External validity. Re-identification is computed on a synthetic identifier universe, but the result is structural—low-entropy enumerable identifiers cannot be anonymized by hashing—and matches the cryptographic literature [8, 5]. The economics model is illustrative and not a pricing recommendation.

19 Discussion

Three themes recur. First, *explainability is a safety feature, not a nicety*: the deterministic layer’s transparency is what makes dispute, reinvestigation, and audit feasible under FCRA. Second, *the network’s strength is also its danger*: cross-employer memory is the unique value, but a false signal in that memory is repeated and amplified, which is why the privacy, incentive, and governance designs are not separable from the detection design. Third, *incentives must reward information, not verdicts*: the single most consequential design error we identify is paying for fraud findings, and it is also the cheapest to fix.

Table 18: Alternatives considered and why the chosen design is preferred.

Alternative	Why attractive	Why rejected
Central trusted clearinghouse	simple matching, low latency	single point of breach; concentrates derogatory PII; weak GDPR posture
ML-only detection	captures subtle patterns	opaque; undefendable in dispute; high FP on atypical genuine candidates
Reputation only (no threshold)	cheap; no MPC	a single confident farmer still injects false flags
Blockchain ledger of flags	tamper-evidence; decentralization	immutable derogatory records conflict with erasure and dispute rights
Deterministic-first + threshold-PSI + symmetric incentives (this work)	explainable, minimally leaky, gaming-resistant	higher cryptographic cost (acceptable off the synchronous path)

20 Ethics and Responsible Disclosure

The dominant risk is not missed fraud but *wrongful suspicion*: a false positive can quietly cost a real person a job, and a shared false positive can do so repeatedly and invisibly. This motivates the advisory-only stance, the dispute mechanism, the false-positive-cost metric, and the evidence-gated network. The work is dual-use—a detector and a generator co-evolve, as Section 10.5 shows—and the inferential signals risk encoding bias against candidates with non-standard or low-digital-footprint histories, including along national and cultural lines. We therefore (a) keep inferential and network signals advisory; (b) prioritize the explainable deterministic layer; (c) release the implementation openly for scrutiny before any commercial use; and (d) defer commercial operation, which triggers the heaviest regulatory duties. We do not claim the deterministic layer is novel in isolation; its value is as the explainable backbone of an integrated, shared, regulation-aware system.

21 Limitations and Research Agenda

The central open item is empirical and external: a multi-organization study with consenting partners to validate the flywheel at scale, and a measurement of the deterministic and inferential false-positive rates on a real résumé corpus with genuine, coherent employment dates and across demographic and cultural cohorts (a fairness audit). As Section 9.1 documents, such a corpus is not currently available publicly—the leading candidates either lack dates or are synthetically contaminated—so this measurement awaits either a suitable public release or a consented deployment; our realistic generated population (Section 10.2) is the present-day stand-in. On the cryptographic side, a formal security proof and a performance evaluation of the chosen threshold-PSI protocol are needed. On the detection side, richer inferential features and their adaptive-adversary robustness deserve study, as does integration with real LLM-generated résumés rather than our stylized fabricator. None of these changes the framework; they instantiate and stress it.

22 Conclusion

Employment-experience fraud is a coordination failure: defenders act alone while attackers reuse fabricated identities across them. Other industries closed this gap with privacy-preserving contributory networks, but doing so for hiring requires jointly solving detection, re-identification-resistant sharing, contribution incentives, and regulatory constraints—problems previously studied apart. We framed that integration; gave a deterministic, explainable detection core; demonstrated, with confidence intervals and parameter sweeps, why two common shortcuts (hash-as-anonymization and fraud-weighted rewards) fail and how to fix them; quantified the contributory flywheel and fraud-ring detection; and grounded every decision in FCRA and GDPR constraints. The next step is an open, multi-organization study to validate the network effect and to measure, above all, the cost the system imposes on the innocent.

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A Notation

Symbol	Meaning
a, g, y	age, graduation year, total claimed years
$J = \{(s_i, e_i, \tau_i)\}$	job intervals (start, end, type)
θ_o, θ_y	overlap (months) and inflation (years) tolerances
t	corroboration / disclosure threshold
$\text{rep}[p]$	reputation weight of participant p
$\mathcal{A}_C, \mathcal{A}_P, \mathcal{A}_I$	candidate, privacy, incentive adversaries

B Flag reference

Flag code	Severity	Points
TIMELINE_IMPOSSIBLE	Critical	+50
JOB_BEFORE_GRADUATION	Critical	+40
EMPLOYMENT_OVERLAP	High	+25 each
EXPERIENCE_INFLATION	High	+25
NETWORK_FLAG(c)	High/Critical	threshold-gated

C Hyperparameters

Experiment	Parameter	Value
Deterministic	population n / fraud rate	3000 / 0.5
Deterministic	θ_o / θ_y	3 mo / 1 yr
Incentive	participants / candidates	12 / 4000
Incentive	fraud prior / borderline	0.20 / 0.30
Incentive	threshold t (proposed)	2
Incentive	seeds	30
Privacy	universe / shared	20000 / 4000
Agent	per class n / seeds	1500 / 10
Flywheel	orgs / apps-per-actor	50 / 4
All	master seed	20260612

D Reproducibility checklist

The released repository [11] contains: the deterministic engine and unit tests; the seed tables and deterministic dataset generator with a provenance hash; the fraud-injection, evaluation, incentive, privacy, agent, network, and sweep modules; the figure-generation scripts; and a download script for the external corpora. Each command in the repository README reproduces a named table or figure in this paper from the same master seed.

E Full experimental results

All values are means over the stated number of seeds with 95% confidence intervals where applicable; they are emitted directly from `results/study.json` in the released harness.

Table 19: Incentive study: network FP, precision, recall vs. strategic participation (30 seeds).

Strat.	Regime	FP	Precision	Recall
0%	naive	0.000 [0.000, 0.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]
0%	proposed	0.000 [0.000, 0.000]	1.000 [1.000, 1.000]	0.669 [0.663, 0.674]
10%	naive	0.040 [0.027, 0.054]	0.875 [0.838, 0.911]	1.000 [1.000, 1.000]
10%	proposed	0.000 [0.000, 0.000]	1.000 [1.000, 1.000]	0.610 [0.591, 0.629]
25%	naive	0.109 [0.090, 0.128]	0.713 [0.674, 0.752]	1.000 [1.000, 1.000]
25%	proposed	0.000 [0.000, 0.000]	1.000 [1.000, 1.000]	0.513 [0.488, 0.539]
40%	naive	0.160 [0.145, 0.176]	0.617 [0.591, 0.643]	1.000 [1.000, 1.000]
40%	proposed	0.000 [0.000, 0.000]	1.000 [1.000, 1.000]	0.448 [0.428, 0.467]
50%	naive	0.188 [0.173, 0.203]	0.577 [0.557, 0.597]	1.000 [1.000, 1.000]
50%	proposed	0.000 [0.000, 0.000]	1.000 [1.000, 1.000]	0.409 [0.391, 0.427]

Table 20: Threshold sweep (proposed regime, 50% strategic, 30 seeds).

t	FP	Recall
1	0.000 [0.000, 0.000]	0.826 [0.807, 0.845]
2	0.000 [0.000, 0.000]	0.409 [0.391, 0.427]
3	0.000 [0.000, 0.000]	0.037 [0.024, 0.049]
4	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]

Table 21: Re-identification vs. adversary coverage (10 seeds).

Coverage	Naive re-id	Keyed (outsider)
10%	0.099 [0.095, 0.102]	0.000 [0.000, 0.000]
25%	0.252 [0.247, 0.257]	0.000 [0.000, 0.000]
50%	0.501 [0.496, 0.506]	0.000 [0.000, 0.000]
75%	0.751 [0.748, 0.754]	0.000 [0.000, 0.000]
100%	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]

Table 22: Regularity-detector AUC vs. fabrication sophistication (10 seeds).

Noise	ROC-AUC
0.0	0.896 [0.891, 0.901]
0.25	0.529 [0.524, 0.534]
0.5	0.304 [0.298, 0.311]
0.75	0.203 [0.195, 0.212]
1.0	0.168 [0.161, 0.176]

Table 23: Single-feature vs combined regularity AUC (noise = 0).

Feature	ROC-AUC
january	0.577
uniformity	0.853
no gap	0.750
roundness	0.577
combined	0.899

F Reproduction map

Each result in this paper is regenerated by a single command in the released harness [11] from master seed 20260612.

G Glossary

Contributory network a shared database to which participants add signals and from which all benefit; the value grows with participation (the flywheel).

Corroboration threshold t the minimum number of independent organizations that must report a candidate before a network concern is disclosed.

Deterministic layer rule-based checks on a candidate’s own claims that fire only on internal contradiction.

Honest-but-curious an adversary that follows the protocol but tries to learn more than it should from what it observes.

Private set intersection (PSI) a protocol letting parties learn their sets’ intersection (or a thresholded function of it) without revealing the rest.

Re-identification rate the fraction of shared candidates an adversary can map back to real identities.

Regularity score a stylometric measure of how “too clean” a trajectory is, used as an inferential signal against naive AI fabrication.

Strategic participant a consortium member that games the reward schedule rather than reporting truthfully.

Table 24: Network flywheel and fraud-ring detection (20 seeds).

Participation	Catch rate
10%	0.011 [0.010, 0.011]
25%	0.064 [0.062, 0.065]
50%	0.250 [0.248, 0.252]
75%	0.461 [0.459, 0.463]
100%	0.667 [0.667, 0.667]
Identities k	Ring detection
1	0.000 [0.000, 0.000]
2	0.493 [0.488, 0.499]
3	0.783 [0.777, 0.788]
4	0.915 [0.912, 0.918]
5	0.967 [0.965, 0.970]
6	0.989 [0.987, 0.991]

Artifact	Module	Command
Table 4	<code>evaluate</code>	<code>run_experiment</code>
Fig. 2, Table 5	<code>sweeps</code>	<code>make_study_figures</code>
Tables 7, 8, Figs. 4a, 5	<code>sim_incentive, sweeps</code>	<code>study</code>
Table 9, Figs. 6a, 6b	<code>sim_privacy, sweeps</code>	<code>study</code>
Table 10, Figs. 7a, 7b	<code>sim_agent, sweeps</code>	<code>study</code>
Fig. 8	<code>sim_network</code>	<code>study</code>
Fig. 9	<code>sim_network</code>	<code>study</code>
Table 11, Fig. 10	<code>sweeps</code>	<code>study</code>
App. tables 19–24	<code>all</code>	<code>study → study.json</code>

H Worked numeric example

Consider a record with age $a = 24$, graduation year $g = 2021$, claimed experience $y = 15$, and two jobs $\{(2019, 2023), (2020, 2024)\}$ both full-time. `TIMELINESCORE` evaluates: $y = 15 > a - 18 = 6$, so `TIMELINE_IMPOSSIBLE` fires (+50); the earliest start $2019 < g = 2021$, so `JOB_BEFORE_GRADUATION` fires (+40); the two intervals overlap by roughly 36 months $> \theta_o$, so `EMPLOYMENT_OVERLAP` fires (+25); and $y = 15$ exceeds the summed tenure $4 + 4 = 8$ by more than θ_y , so `EXPERIENCE_INFLATION` fires (+25). The raw total is 140, capped at 100. Every flag carries a human-legible reason, so the verdict is fully explainable and reproducible—the same input always yields the same output, which is what makes FCRA reinvestigation tractable.

I Per-fraud-type detection detail

Injected fraud type	Triggering flag(s)	Recall
Impossible experience	<code>TIMELINE_IMPOSSIBLE</code> (+ inflation)	1.00
Job before graduation	<code>JOB_BEFORE_GRADUATION</code>	1.00
Employment overlap	<code>EMPLOYMENT_OVERLAP</code>	1.00
Experience inflation	<code>EXPERIENCE_INFLATION</code> (+ impossible 0.29)	1.00
AI-fabricated (consistent)	none (deterministic)	0.00

J Deployment roadmap

Phase	Goal	Gate
1. Open release	scrutiny of method & code	community review
2. Pilot consortium	validate flywheel with consenting orgs	data-processing agreements
3. Fairness audit	FP rate across cohorts on KARRIEREWEGE	published audit
4. PSI hardening	formal proof + performance	security review
5. Regulated operation	advisory product under FCRA/GDPR	legal counsel sign-off

The ordering is deliberate: each gate that protects candidates (scrutiny, fairness audit, security review, legal sign-off) precedes the step that could harm them (operation). Commercial deployment is last, not first.

K Extended ethics: bias vectors

The inferential layer’s signals carry identifiable bias vectors that a deployment must monitor: digital-footprint signals disadvantage privacy-conscious and older candidates; web-presence signals disadvantage candidates with common names and those outside English-language media; employer-registry signals disadvantage those from regions with poor registry coverage; and “regularity” signals can disadvantage genuinely linear careers. None of these is a reason to abandon the signals, but each is a reason to keep them advisory, to measure false-positive rates per cohort before relying on them, and to give candidates a dispute path. The deterministic layer is comparatively bias-resistant because it tests only internal logical consistency, but it is not bias-free: graduation-date rules can misfire on non-traditional education timelines, which is why its tolerances are explicit and auditable.

L Co-observation and threshold selection

The threshold t should be chosen jointly with expected co-observation, which itself rises with participation. If each repeat actor applies to m organizations drawn from N , of which a fraction π participate, the number of in-network observations of a given actor is approximately $\text{Binomial}(m, \pi)$, so the probability of reaching threshold t is $\Pr[X \geq t]$ with $X \sim \text{Binomial}(m, \pi)$. This is why the flywheel of Section 10.6 is convex in π : doubling participation more than doubles the chance of t -fold corroboration once π exceeds t/m . In practice t is set to the smallest value that keeps the network false-positive rate negligible at the current π , and is lowered as participation grows—a managed trade between safety (high t) and recall (low t) that the operator tunes against measured co-observation rather than fixing once.

M Security argument (sketch)

We sketch why the threshold-PSI construction meets goal G2 of Property 1. The underlying over-threshold multi-party PSI reveals, for a queried identifier, only whether it appears in at least t parties’ sets, and the count if so [5]; it reveals nothing for sub-threshold identifiers and does not attribute matches to specific organizations. An honest-but-curious participant therefore learns membership only for candidates already corroborated by t organizations—a set that, by construction, it can only expand by contributing its own genuine observations, not by offline enumeration, because the protocol provides no oracle to test arbitrary guesses without participation. This is the property

that the naive and keyed constructions lack: both expose an offline-testable function of the identifier, which a low-entropy input space renders equivalent to publication. A full proof requires instantiating Π and is deferred to the cryptographic agenda of Section 21.

Data and artifact availability

The implementation, evaluation harness, seed tables, generated datasets, figure scripts, and the exact commands that reproduce every table and figure are released as open source [11]. External corpora (KARRIEREWEGE, ResumeAtlas, Kaggle) are obtained via the included download script and are not redistributed. All randomized results use master seed 20260612 and report confidence intervals over the stated number of seeds.